

S V V Sai Sri Vallabh Pataneni

Research Assistant/Associate (PhD): Fuel Cell Systems, PEM & SOFC

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SUMMARY

An Automotive Engineering Master's student at RWTH Aachen with over two years of hands-on PEM fuel-cell systems experience, from a 1 kW student-competition stack to a 100 kW, 1.5 t FCEV pickup truck at Robert Bosch to a 150-200 kW aircraft powertrain at DLR. I design fuel-cell system control architectures and 0D lumped-parameter plant models in MATLAB/Simulink, and I run structured test-bench campaigns and correlate them against measurement data. Seeking a Research Assistant/Associate PhD position at RWTH's Chair of Thermodynamics of Mobile Energy Conversion Systems (TME) to bring this system-level modeling, control and test-bench experience to fuel-cell systems research for mobile applications.

KEY SKILLS

Fuel-cell systems & controls

PEM fuel-cell system architecture OD lumped-parameter plant modeling feed-forward/PI/decoupler control design
back-pressure-valve (BPV) protection control operating-strategy optimization balance-of-plant CAN communication

Modeling & simulation

MATLAB/Simulink Genetic Algorithms Gradient Descent DP/ECMS benchmarking
thermal-management digital twins

Test & measurement

fuel-cell test-bench build & commissioning single-cell & stack testing polarisation curves purging/humidity cycling
measurement-data evaluation & troubleshooting

Programming & data

Python (NumPy, Pandas) C/C++ Power BI

Design

CATIA V5 Siemens NX

PROFESSIONAL EXPERIENCE

Werkstudent, Fuel-Cell Systems (Aircraft), DLR Institute for Small Aircraft Technologies (INK)

01/2026 – Present

Würselen (Aachen), Germany

- **Cathode-Side Control Architecture:** Designing the cathode-side control architecture for the ~150-200 kW fuel-cell system powering EnginePod, DLR's hydrogen-electric powertrain for a 9-seater commuter aircraft, starting with an extensive 0D lumped-parameter plant model of the full cathode loop in MATLAB/Simulink.
- **Control Design:** Architected feed-forward, PI and decoupler control for cathode-side operation and back-pressure-valve (BPV) protection, now tuning control parameters against real system measurement data.
- **Thermal-Management Digital Twin:** Built a MATLAB/Simulink digital twin of the complete thermal-management system to derive design specifications (coolant flow, temperature range, coolant pressure), driving procurement of the coolant pump and heat exchanger.
- **Component Selection & System Rig:** Selected every thermal-management component against the derived specifications and designed a compact, aircraft-engine-sized fuel-cell system rig in CATIA V5, optimized for easy transferability.

Mechanical / Suspension Team Lead & Pit Crew, Ecogenium e.V. (RWTH student team)

06/2024 – Present

Aachen, Germany

- **Fuel-Cell System Testing & Troubleshooting:** Leading this season's fuel-cell system testing, integration and fault diagnosis on our newly built water-cooled fuel-cell system, running structured test campaigns across the electrical, thermal and fluidic subsystems.
- **Energy-Management & Cooling System Design:** Designed this season's energy-management and cooling-system control strategy for the water-cooled fuel-cell system in MATLAB/Simulink.
- **Fuel-Cell Test Bench:** Engineered, fabricated and electronically integrated a standalone fuel-cell test bench for the prior air-cooled stack, running purging, humidity cycling, short-circuiting and polarisation-curve experiments and mapping its optimal thermal operating window.
- **End-to-End System Ownership:** Holistic ownership of fuel-cell system setup, from component sizing and selection through CAN-bus communication and control design to test and experimentation, gained across a runner-up finish at Shell Eco-Marathon 2025.

Operating Strategy / System Simulation Intern, Robert Bosch GmbH

09/2025 – 12/2025

Schwieberdingen, Germany

- **Operating-Strategy Modelling:** Developed and validated an iterative Simulink model of a rule-based power-split operating strategy for a 100 kW-stack, 1.5 t FCEV pickup-truck platform in the (FC)EVsim PHEV module, correlating outputs against reference Excel and measurement data for high model fidelity.
- **Multi-Objective Optimization:** Executed multi-variable parametric optimization using Genetic Algorithms and Gradient Descent to generate control MAPs for charge sustenance and minimal fuel consumption, benchmarked against Dynamic Programming and ECMS.
- **Analysis & Documentation:** Built post-processing and KPI analysis (power distribution, SOC window, efficiency, degradation) and co-authored a peer-reviewed paper on the rule-based FCEV control strategy.

EDUCATION

M.Sc. Automotive Engineering, RWTH Aachen University

09/2023 – present

Aachen, Germany · GPA 1.7 (1.2 in core automotive courses) · Master's thesis in progress at DLR: fuel-cell system cathode-side control architecture · Relevant modules: Fuel-Cell System Technology, Alternative/Mobile Propulsion, Simulation Sciences, FEM, CFD

B.Tech Mechanical Engineering, JNTU Hyderabad

06/2019 – 04/2023

Hyderabad, India · CGPA 9.05 (GPA 1.3, German scale)

PUBLICATIONS

- Co-author, peer-reviewed paper on a rule-based operating-strategy control approach for FCEV powertrains (Robert Bosch GmbH), in publication.

LANGUAGES

English (Fluent, C1) · German (A2-B1) · Telugu / Hindi (Native)